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Github Repository:
https://github.com/Keane-Lim/FinalProj

SPACE X FALCON 9 FIRST STAGE LANDING PREDICTION

EXECUTIVE SUMMARY

- Project Objective: Predict whether the Falcon 9 first stage will land successfully, which determines the cost of a launch ($62M vs $165M+).
- Methodology:
  - Gathered data via SpaceX API and Wikipedia Web Scraping.
  - Performed Data Wrangling and EDA using SQL and Python (Pandas/Matplotlib).
  - Built interactive maps with Folium and a dashboard with Plotly Dash.
  - Trained and tuned 4 Machine Learning classification models.
- Key Finding: The Logistic Regression Model achieved the highest test accuracy of 83.3%, making it the optimal model for launch cost risk assessment.

INTRODUCTION AND PROBLEM STATEMENT

- **Background:**
SpaceX launches Falcon 9 at $62m per flight, while traditional aerospace costs $165m+. Cost efficiency is driven entirely by first-stage booster reusability.

- **Problem:**
Predict whether a booster will land successfully based on variables like payload mass, target orbit, and launch site, allowing competitors to quote prices more intelligently

DATA COLLECTION PIPELINE

A. SpaceX API Integration:

- Requested primary endpoints from the SpaceX API (/v4/launches/past).
- Unpacked highly nested JSON lists for rocket cores, payloads, and launchpads.
- Structured features into a localized Pandas DataFrame named spacex_df.

B. Wikipedia Web Scraping:

- Sent automated HTTP requests to Wikipedia's Falcon 9 launch list page.
- Parsed HTML data tables structurally via BeautifulSoup.
- Extracted critical records: Launch Site, Payload Mass, Orbit Type, and Landing Outcomes from unstructured text blocks.

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DATA WRANGLING METHODOLOGY

To ensure full data integrity before training models, missing values in the Payload Mass (kg) column were imputed using the column mean.

We extracted target variables to generate explicit binary labels: 1 for a Successful Recovery (e.g., True ASDS, True RTLS) and 0 for an Unsuccessful Outcome (e.g., False ASDS, False RTLS, Ocean/Water crashes).

EXPLORATORY DATA ANALYSIS (EDA)

Visual Insights:

Flight Number vs. Payload Mass: Later flight numbers exhibit significantly higher success rates, indicating technological improvements over time.

Success Rate by Orbit Type: High-altitude or specific orbits like ES-L1, GEO, HEO, and SSO achieve a perfect 100% success rate, whereas GTO landing success fluctuates around 60%.

SQL ANALYTICAL QUERIES

STRUCTURED QUERY LANGUAGE WAS UTILIZED TO RUN DEEP INSPECTIONS ON THE DATABASE TABLES:

FORMULATED QUERIES TO FILTER AND LOCATE ALL UNIQUE LAUNCH SITES: CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40.

Interactive Geospatial Tracking (Folium)

Geospatial analysis was conducted using the Folium library to chart launch pad coordinates:

1. Rendered interactive map pins and clustered active launch zones utilizing MarkerClusters.

2. Proximity Distance Analysis: Computed the shortest geodesic distance from each launchpad to nearby commercial coastlines, national highways, and active railroads using Folium PolyLines.

Key Insight: All pads are intentionally built immediately adjacent to ocean shorelines to mitigate land-based destruction risks during potential rocket failures.

INTERACTIVE LIVE DASHBOARD (PLOTLY DASH)

A SCALABLE INTERACTIVE DASHBOARD WAS DEVELOPED TO ALLOW NON-TECHNICAL STAKEHOLDERS TO SLICE METRICS:

EMBEDDED AN INTERACTIVE DCC.DROPDOWN SELECTION COMPONENT ALLOWING INSTANT SWITCHING BETWEEN GLOBAL AGGREGATION VIEWS AND SPECIFIC LAUNCH PAD SEGMENTS.

CONNECTED A RESPONSIVE DCC. RANGESLIDER FILTERING PRECISE PAYLOAD MASS BANDS, WHICH DYNAMICALLY REFRESHES SUCCESS RATIOS AND CORE SCATTER CHARTS ON THE FLY.

EVALUATION RESULTS & FINAL COMPARISON

Classification Model	Train Accuracy	Test Accuracy
Logistic Regression	84.6%	87.5%
SVM	84.8%	83.3%
Decision trees	88.8%	77.8%
KNN	84.8%	83.3%